

MAT282 Intermediate Statistics: Midterm Exam

Franklin University Switzerland · Fall 2026 · Instructor: Assoc. Prof. C. Burke · Oct 9, 2026

Name: _____ Form: A Score: _____ / 50

Time: about 45 minutes · **Allowed:** pen/pencil, calculator, ruler · **Total:** 50 points · Show working for every calculation: marks are for method as well as answers. Give numerical answers to 2 decimal places unless told otherwise. The data are **synthetic teaching data** from a fictional seaside gelato café, not real records.

ANSWER KEY / MARKING NOTES — instructor copy. Do not distribute.

Scenario. A seaside café recorded daily sales and weather for 168 trading days (7 April–21 September 2025). You will work with a **13-day sample** of those days. The manager's belief: *"Hot days are good for business."* Your tasks follow the way we work in class: state ideas that can be tested, describe the data, test the relationship, compare periods, and read model output.

Operationalisation. **Temperature** = daily mean outdoor temperature (°C) during opening hours. **Gelato Units** = number of standard servings of gelato sold that day.

Part 1 · Concepts: hypothesis, falsification, data quality (8 points)

1. Hypothesis & falsification [LO 1]

[4]

The manager's belief is vague. (a) Rewrite it as a **testable hypothesis** linking temperature to gelato units, including the **direction** of the relationship and the statistic you would use. (2) (b) Describe a specific result from the 13-day sample that would **falsify** your hypothesis. (2)

Key: (a) "As daily mean temperature rises, {yn} sold rises (positive correlation, $r > 0$)"; direction + statistic = 1 each. (b) Any specific falsifying result, e.g. $r \leq 0$ or a clearly flat/negative scatter; statement that a hypothesis cannot be "proved" earns credit but is not required.

2. Validity & reliability [LO 2]

[4]

(a) "Daily mean temperature during opening hours" is used to capture *how hot it felt to customers when deciding to buy gelato*. Give **one reason** this operationalisation might be a weak (invalid) measure of that idea. (2) (b) Name **one thing** that could make the temperature *or* sales measurement unreliable, and say what it would do to the data. (2)

Key: (a) e.g. daily mean hides peaks (afternoon heat), ignores humidity/wind/"feels like", measured at one station. (b) e.g. sensor drift/mis-calibration, counting errors, inconsistent serving size: adds noise or systematic bias. 1 pt for the issue, 1 pt for explanation each.

Part 2 · Describing the data: calculation and visualisation (13 points)

Sample data (13 days). Sum of gelato units = 2431.

Day	1	2	3	4	5	6	7	8	9	10	11	12	13
Date	Apr 7	May 5	May 23	May 28	Jun 16	Jul 30	Aug 2	Aug 16	Aug 17	Aug 23	Aug 28	Sep 2	Sep 16
Temp (°C)	19.7	20.2	19.3	12.6	17.7	22.7	30.9	22.9	21	25.5	21.7	28.2	26.2

Gelato Units	166	225	164	111	131	134	294	193	209	198	196	194	216
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3. Mean & median [LO 3]

[3]

Calculate the **mean** and **median** of gelato units. Which is the better summary of a “typical day” here, and why?

Key: Mean = $2431/13 = 187$; median (7th of 13 ordered) = 194. Mean pulled toward the outlier (294); median is more resistant. 1+1+1.

4. Quartiles & outliers [LO 3, 4]

[4]

Sort the gelato units values. Using the Sheets convention for 13 values (**Q1 = 4th** and **Q3 = 10th** ordered value), find Q1, Q3 and the IQR, calculate the **1.5×IQR fences**, and state which day(s), if any, are outliers.

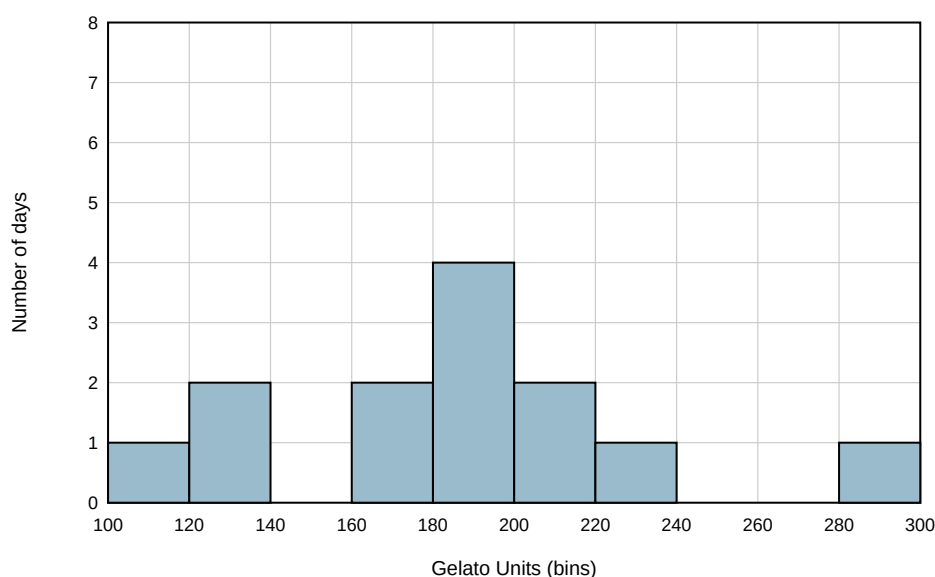
Key: Sorted: [111, 131, 134, 164, 166, 193, 194, 196, 198, 209, 216, 225, 294]. Q1=164, Q3=209, IQR=45; fences 96.5 and 276.5. Outlier: 294 units on Aug 2 (temp 30.9 °C). Marks: Q1/Q3 (1), IQR+fences (2), identification (1).

5. Histogram [LO 4]

[3]

Draw a **histogram** of gelato units using the bins marked on the axis (each bin includes its lower edge). Label the vertical axis as a frequency count, and then describe the **shape** in one sentence.

Key: Bars: 100–120: 1, 120–140: 2, 140–160: 0, 160–180: 2, 180–200: 4, 200–220: 2, 220–240: 1, 240–260: 0, 260–280: 0, 280–300: 1. Shape: roughly symmetric/slightly right-skewed with one high value separated from the rest. 2 pts for correct bars (allow one slip), 1 for shape.



6. Spread & Sheets [LO 3, 5]

[3]

The sum of squared deviations is $\sum(y - \bar{y})^2 = 27196$. (a) Calculate the **sample standard deviation**. (2) (b) Write the Sheets formula you would use if the gelato units sit in cells C2:C14. (1)

Key: (a) $\sqrt{27196/12} = 47.61$. (b) =STDEV.S(C2:C14) (or STDEV); STDEV.P/STDEVP gets 0. Using n instead of n-1 loses 1. z of outlier = 2.25.

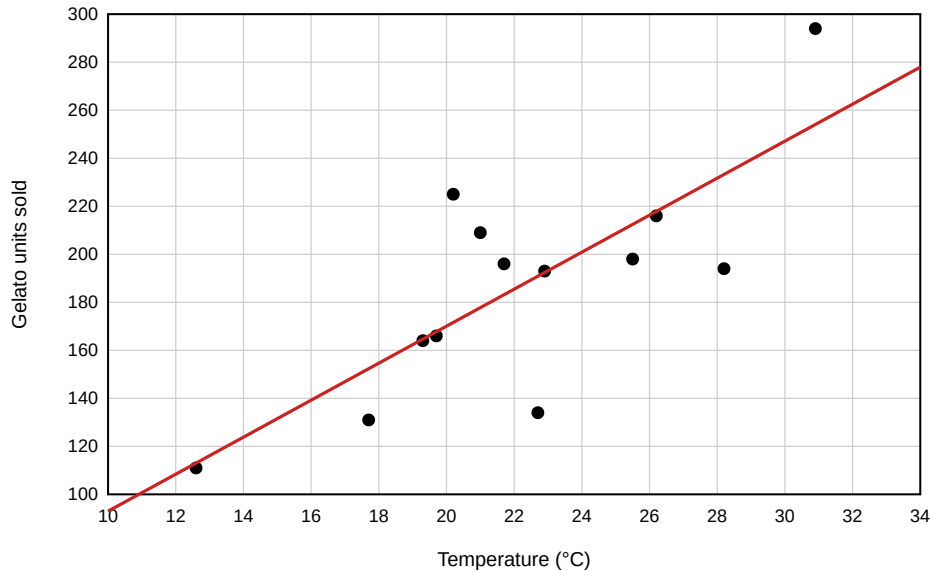
Part 3 · Relationship and prediction (13 points)

7. Scatterplot [LO 4]

[3]

Plot the 13 days (temperature on the x-axis, gelato units on the y-axis) and draw a **line of best fit by eye**. Circle the outlier you found in Question 4.

Key: All 13 points within one small square; line has positive slope passing through (mean x, mean y)=(22.2, 187). Computed line $\hat{y} = 15.91 + 7.71x$. 2 pts points, 1 pt line + circle.



8. Correlation [LO 3, 6]

[2]

Estimate the correlation coefficient r for your scatterplot (no calculation). Explain in one sentence what the **sign** and the **size** of r tell you.

Key: Computed $r = 0.77$; accept ± 0.15 of that. Sign = direction; size = strength/tightness around a line. 1+1.

9. Prediction & error [LO 5]

[4]

A line fitted to these data is $\hat{y} = 15.91 + 7.71 \cdot \text{temperature}$. (a) Write the Sheets formula *FORECAST* to predict gelato units for a 24 °C day, if temperatures are in B2:B14 and gelato units are in C2:C14, and calculate the prediction. (2) (b) For the two days below, calculate the residual (actual – predicted) and the **mean absolute deviation** (MAE) of the two predictions. (2)

Day	Temp	Actual
May 23	19.3	164
Aug 17	21	209

Key: (a) =FORECAST(24,C2:C14,B2:B14) — argument order: x, known_y, known_x (1); prediction 200.95 (1). (b) May 23: predicted 164.71, residual -0.71, Aug 17: predicted 177.82, residual 31.18; MAE = 15.95. Allow ECF from (a) rounding.

10. Interpretation & extrapolation [LO 6, 7]**[2]**

(a) Interpret the slope in one sentence **in context**. (1) (b) Would you use the line to predict gelato units on a 15 °C day? Why or why not? (1)

Key: (a) Each +1 °C is associated with about 7.71 more servings of gelato per day on average (association, not proof of cause). (b) 15 °C is outside the sampled range (12.6–30.9 °C) → extrapolation, and the predicted value (131.56) may be unreliable.

11. Weighted average [LO 5]**[2]**

Over all 168 days, the café grouped days by temperature. Calculate the **weighted mean** of daily gelato units across the three bands, and say why it differs from the simple average of the three band means.

Band	Days (n)	Mean units
below 20 °C	34	160.8
20 to 25 °C	89	189.7
above 25 °C	45	218.1

Key: Weighted mean = $\Sigma(n \cdot \text{mean}) / \Sigma n = 191.4$ (1); simple average of band means = 189.5; bands have different numbers of days, so each must be weighted by its n (1).

Part 4 · Comparing periods and significance (6 points)**12. Difference of means [LO 3, 6, 7]****[6]**

On 30 June 2025 the café raised the gelato price from €4.00 to €4.50. Daily gelato sales (full data):

	Days (n)	Mean units	Std dev
Before (to 30 Jun)	85	183.2	36.3
After (from 1 Jul)	83	199.9	33.5

(a) Calculate the **standard error** of the difference in means, $SE = \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$. (2) (b) Calculate $t = (\text{difference})/SE$, and decide whether the difference is statistically significant at 5% (use $|t| > 1.97$ as the cutoff). (2) (c) Is it sensible to conclude that the price change *caused* the change in sales? Give one rival explanation. (2)

Key: (a) $SE = 5.39$ (1 for formula, 1 for value). (b) $\text{diff} = 16.73$; $t = 3.10$; $|t| > 1.97 \rightarrow$ significant (1+1). (c) No: observational before/after; rival = season (post period is the hotter summer), events, weekdays; also significance $\neq$ importance. 1 for “not causal”, 1 for specific rival.

Part 5 · Reading model output (LINEST) (6 points)

13. Computation: interpreting LINEST [LO 5, 6, 7]

[6]

A student fits a multiple regression for **gelato_units** on **temperature_c**, **beach_event**, **weekend** (in that column order) with `=LINEST(y, X, TRUE, TRUE)` on all 168 days. Sheets returns (rounded):

	Col 1	Col 2	Col 3	Col 4
Row 1	34.16	25.74	5.57	48.97
Row 2	3.85	4.27	0.52	11.86
Row 3	0.61	22.52	#N/A	#N/A
Row 4	86.39	164.00	#N/A	#N/A
Row 5	131402.67	83146.74	#N/A	#N/A

Row 1: coefficients (last X column first, intercept last) · Row 2: standard errors · Row 3: R^2 , SE of y · Row 4: F, df · Row 5: SS regression, SS residual.

(a) Which column holds the **temperature** coefficient? State its value and interpret it **holding the other variables constant**. (2)

(b) Calculate the t-statistic for **beach_event** and say whether it is significant (cutoff $|t| > 1.97$). Interpret the coefficient in plain language. (2)

(c) What does $R^2 = 0.61$ mean here? Using a simple regression with temperature alone, $R^2 = 0.34$. What does the increase suggest? (2)

Key: (a) Column 3 (coefficients are reversed: weekend, event, temp, intercept) = 5.57: each +1 °C goes with about 5.57 more units, holding event and weekend constant (1+1). (b) $t = 25.74/4.27 = 6.03 \rightarrow$ significant; event days sell about 25.7 more units than non-event days, other things constant (1+1). (c) Model explains 61.2% of the variation in daily units; adding event/weekend raises R^2 from 0.34 to 0.61, so they carry extra information beyond temperature (1+1). Weekend coefficient 34.2; intercept 49.0.

Part 6 · Design your own test (4 points)

Codebook for all variables in the full data set:

Variable	Meaning / operationalisation
temperature_c	Daily mean outdoor temperature (°C) during opening hours
sunshine_hours, rainfall_mm, wind_kmh, humidity_pct	Daily weather during opening hours
beach_event	1 if an organised public event occurred on the nearby beach that day, else 0
weekend	1 if Saturday or Sunday, else 0
gelato / iced coffee / pastry price	Posted price in € for one standard serving; changed on 30 June 2025 and varied by daily price trials
gelato / iced coffee / pastry units	Number of standard servings sold that day (no stock-outs or closures)

14. *Blueprint your own analysis [LO 1, 5, 7]*

[4]

Choose **any two variables** from the codebook that we have *not* examined in this exam. In about 5 sentences, give: (i) your **hypothesis** and why you expect it; (ii) your **expected finding** (direction and rough strength); (iii) the **analysis** you would run in Sheets (columns, function or chart); (iv) a result that would **falsify** you. Creativity is welcome: reasoning matters more than being right.

Key: Rubric (1 each): (i) clear, testable hypothesis with a reason; (ii) specific expected direction/size; (iii) feasible Sheets analysis that matches variable types (e.g. CORREL, scatter, AVERAGEIF, t-test, LINEST); (iv) concrete falsifying result. Accept any sensible pair (e.g. rain → coffee, wind → gelato, price → units, weekend → pastry). Do not reward only “correlation ≠ causation”.